

Blueprint Arsitektur: Central Analytics API (App B) — FINAL v2

Ekosistem Agregator Musik — ClickHouse Analytics Engine

Status: ✔ Semua keputusan arsitektur terkonfirmasi

Arsitektur Sistem

```
graph TB
    subgraph "Satu VM"
        subgraph "App A – NeonVault (port 3000)"
            A_UI["UI (Dashboard + Admin Upload)"]
            A_BE["Next.js Backend"]
            A_DB["PostgreSQL"]
        end
        subgraph "App B – Analytics API (localhost:3001)"
            B_API["Fastify REST API"]
            B_ETL["ETL Worker"]
            B_CH["ClickHouse<br/>db_cms_* (dinamis)"]
        end
    end
    A_UI --> A_BE -->|"localhost:3001"| B_API --> B_CH
    A_BE -->|"upload file"| B_ETL --> B_CH
    INET["🌐 Internet"] --> A_UI
    INET -. ->|"🚫 BLOCKED"| B_API
```

Pemetaan Kolom (Sumber Kebenaran)

[!IMPORTANT] *Revenue, views, dan identifier (asset_id, isrc, upc) selalu dari Claim Raw.*

Kolom	Sumber	Key Lookup
report_type, label_source	Diisi saat ingestion	—
adjustment_type, day, country	① Claim Raw	—
video_id, channel_id	① Claim Raw	—
asset_id, asset_channel_id, asset_type	① Claim Raw	—
content_type, policy, claim_type, claim_origin	① Claim Raw	—
custom_id, isrc, upc, grid	① Claim Raw	—
owned_views + 10 kolom revenue	① Claim Raw	—
video_title, video_duration_sec	② Claim Summary	by video_id
username, uploader, channel_display_name	② Claim Summary	by video_id

multiple_claims, category	② Claim Summary	by video_id
asset_labels	② Claim Summary	by video_id
artist, asset_title, album, label	③ Asset Summary	by isrc

Jika ISRC kosong → metadata musik (artist, album, label) **dibiarkan kosong**. Tidak ada fallback. `Label` = nama label (Sony, Warner). `asset_Labels` = marker internal CMS grouping kepemilikan.

Tabel ClickHouse

ads_revenue_enriched

```
CREATE TABLE IF NOT EXISTS ads_revenue_enriched (
  -- Metadata
  report_type          LowCardinality(String) DEFAULT 'claim_raw'
                        COMMENT 'claim_raw | ads_adjustment | sub_adjustment',
  label_source         LowCardinality(String) DEFAULT 'report'
                        COMMENT 'report | reconciled | corrected',
  -- Dari Claim Raw (master)
  adjustment_type      LowCardinality(String) DEFAULT '',
  day                  Date,
  country               LowCardinality(String),
  video_id              String,
  channel_id            String,
  asset_id              String,
  asset_channel_id      String          DEFAULT '',
  asset_type            LowCardinality(String),
  content_type          LowCardinality(String),
  policy                LowCardinality(String),
  claim_type            LowCardinality(String),
  claim_origin          LowCardinality(String),
  custom_id             String          DEFAULT '',
  isrc                  String          DEFAULT '',
  grid                  String          DEFAULT '',
  upc                   String          DEFAULT '',
  -- Enriched dari Claim Summary (by video_id)
  video_title           String          DEFAULT '',
  video_duration_sec    UInt32          DEFAULT 0,
  username              String          DEFAULT '',
  uploader              String          DEFAULT '',
  channel_display_name  LowCardinality(String) DEFAULT '',
  multiple_claims       LowCardinality(String) DEFAULT '',
  category              LowCardinality(String) DEFAULT '',
  asset_labels          String          DEFAULT ''
                        COMMENT 'Marker internal CMS untuk grouping kepemilikan',
  -- Enriched dari Asset Summary (by isrc)
  artist                String          DEFAULT '',
  asset_title           String          DEFAULT '',
  album                 String          DEFAULT '',
  label                 LowCardinality(String) DEFAULT ''
)
```

```

COMMENT 'Nama label/publisher (Sony, Warner)',
-- Revenue (selalu dari Claim Raw)
owned_views                UInt64          DEFAULT 0,
yt_rev_auction              Decimal64(6)    DEFAULT 0,
yt_rev_reserved            Decimal64(6)    DEFAULT 0,
yt_rev_partner_sold_yt_served Decimal64(6)    DEFAULT 0,
yt_rev_partner_sold_p_served Decimal64(6)    DEFAULT 0,
yt_rev_total               Decimal64(6)    DEFAULT 0,
partner_rev_auction        Decimal64(6)    DEFAULT 0,
partner_rev_reserved       Decimal64(6)    DEFAULT 0,
partner_rev_partner_sold_yt_served Decimal64(6)    DEFAULT 0,
partner_rev_partner_sold_p_served Decimal64(6)    DEFAULT 0,
partner_rev_total          Decimal64(6)    DEFAULT 0,
ingested_at               DateTime        DEFAULT now()
)
ENGINE = ReplacingMergeTree(ingested_at)
PARTITION BY toYYYYMM(day)
ORDER BY (day, report_type, asset_id, video_id, channel_id, country, claim_type)
SETTINGS index_granularity = 8192;

```

subscription_revenue

```

CREATE TABLE IF NOT EXISTS subscription_revenue (
  adjustment_type      LowCardinality(String) DEFAULT '',
  day                  Date,
  country              LowCardinality(String),
  video_id             String,
  video_channel_id     String          DEFAULT '',
  asset_id             String,
  asset_channel_id     String          DEFAULT '',
  asset_title          String          DEFAULT '',
  asset_labels         String          DEFAULT ''
  COMMENT 'Marker internal CMS',
  asset_type           LowCardinality(String),
  custom_id            String          DEFAULT '',
  isrc                 String          DEFAULT '',
  upc                  String          DEFAULT '',
  grid                 String          DEFAULT '',
  artist               String          DEFAULT '',
  album                String          DEFAULT '',
  label                LowCardinality(String) DEFAULT ''
  COMMENT 'Nama label/publisher',
  claim_type           LowCardinality(String),
  content_type         LowCardinality(String),
  offer                LowCardinality(String) DEFAULT '',
  owned_views          UInt64          DEFAULT 0,
  monetized_views_audio UInt64          DEFAULT 0,
  monetized_views_audiovisual UInt64      DEFAULT 0,
  monetized_views_total UInt64          DEFAULT 0,
  yt_rev_total         Decimal64(6)    DEFAULT 0,
  partner_rev_pro_rata Decimal64(6)    DEFAULT 0,

```

```

    partner_rev_per_sub_min      Decimal64(6) DEFAULT 0,
    partner_rev_total            Decimal64(6) DEFAULT 0,
    ingested_at                  DateTime      DEFAULT now()
) ENGINE = MergeTree()
PARTITION BY toYYYYMM(day)
ORDER BY (day, asset_id, video_id, country);

```

Tabel Pendukung

```

-- Registrasi CMS dinamis (di default DB)
CREATE TABLE cms_registry (
    cms_id String, cms_name String, db_name String,
    api_key_hash String, is_active UInt8 DEFAULT 1,
    created_at DateTime DEFAULT now()
) ENGINE = ReplacingMergeTree(created_at) ORDER BY (cms_id);

-- Channel → label mapping (per CMS database)
CREATE TABLE channel_label_map (
    channel_id String, asset_label String,
    cms_name LowCardinality(String), notes String DEFAULT '',
    updated_at DateTime DEFAULT now()
) ENGINE = ReplacingMergeTree(updated_at) ORDER BY (channel_id);

-- Job tracking (per CMS database)
CREATE TABLE ingestion_jobs (
    job_id      String,
    job_type     LowCardinality(String)
                COMMENT 'ads_revenue | subscription | reconciliation',
    status       LowCardinality(String) DEFAULT 'pending'
                COMMENT 'pending | processing | completed | failed',
    month        UInt32 COMMENT 'YYYYMM format, e.g. 202601',
    total_rows   UInt64  DEFAULT 0,
    processed_rows UInt64 DEFAULT 0,
    error_message String DEFAULT '',
    started_at   DateTime DEFAULT now(),
    completed_at Nullable(DateTime)
) ENGINE = MergeTree()
ORDER BY (job_id);

-- Unified view
CREATE VIEW v_unified_revenue AS
SELECT day, video_id, asset_id, country,
       'ads' AS revenue_source, report_type AS sub_source,
       owned_views, yt_rev_total, partner_rev_total
FROM ads_revenue_enriched
UNION ALL
SELECT day, video_id, asset_id, country,
       'subscription', 'subscription',
       owned_views, yt_rev_total, partner_rev_total
FROM subscription_revenue;

```

```
-- Materialized View: pre-agregasi harian
CREATE MATERIALIZED VIEW mv_daily_summary
ENGINE = SummingMergeTree()
ORDER BY (day, report_type)
AS SELECT
    day,
    report_type,
    count() AS total_rows,
    sum(owned_views) AS total_views,
    sum(yt_rev_total) AS total_yt_revenue,
    sum(partner_rev_total) AS total_partner_revenue
FROM ads_revenue_enriched
GROUP BY day, report_type;
```

ETL Pipeline & Error Handling

Alur Ingestion

Admin upload via App A:

1. App A forward file ke App B
2. App B buat job di ingestion_jobs (status='pending')
3. DROP PARTITION bulan X (bersihkan data lama + adjustment)
4. Parse staging: videoclaim → Map, asset_summary → Map
5. Stream claim_raw → enrich → batch INSERT 50K rows
6. (opsional) Stream adjustment → enrich → INSERT
7. Auto reconciliation (fix null labels)
8. OPTIMIZE TABLE FINAL
9. Update job status → 'completed'

Error Handling: "Clean Slate"

```
graph TD
    START["Job dimulai"] --> DROP["DROP PARTITION bulan X"]
    DROP --> PARSE["Parse staging files"]
    PARSE -->|Gagal| FAIL
    PARSE -->|OK| STREAM["Stream + enrich + INSERT"]
    STREAM -->|Gagal baris ke-N| CLEANUP["DROP PARTITION bulan X<br/>(buang data parsial)"]
    CLEANUP --> FAIL["Mark job 'failed'<br/>Log error message"]
    STREAM -->|Semua OK| RECON["Reconciliation"]
    RECON --> DONE["Mark job 'completed' ✅"]
    FAIL --> NOTIFY["Admin lihat error<br/>di Job Monitor"]
    NOTIFY --> REUP["Fix file → Upload ulang"]

    style FAIL fill:#dc2626,color:#fff
    style DONE fill:#059669,color:#fff
    style CLEANUP fill:#f59e0b,color:#000
```

Tidak ada retry otomatis. Upload bulanan: admin perlu tahu penyebab error sebelum coba lagi.

Admin Features

Fitur	Fungsi
Channel Label Mapping	Daftarkan rollup channel → fix null labels
Asset Label Correction	Fix label salah (per asset_id / bulk CSV)
Job Monitor	Progress ingestion, error logs
CMS Registry	Tambah/hapus CMS (auto-create DB, user, role)

API Endpoints Fase 1

Method	Path	Fungsi
GET	/api/v1/revenue/daily?month=	Revenue harian (line chart)
GET	/api/v1/views/daily?month=	Views harian (line chart)
GET	/api/v1/analytics/top-assets?month=	Top lagu by revenue
GET	/api/v1/analytics/top-videos?month=	Top video by revenue
GET	/api/v1/analytics/top-channels?month=	Top channel by revenue
GET	/api/v1/analytics/by-country?month=	Sebaran per negara
GET	/api/v1/analytics/by-label?month=	Revenue per label
POST	/api/v1/ingest/ads-revenue	Upload 3-5 file → ETL
POST	/api/v1/ingest/subscription	Upload subscription
GET	/api/v1/ingest/jobs/:id	Status job
CRUD	/api/v1/admin/channel-map	Channel label mapping
CRUD	/api/v1/admin/cms-registry	Registrasi CMS
POST	/api/v1/admin/correct-label	Koreksi label

Cross-CMS & Music Video Tracking

Fase 1: Query by ISRC / UPC — MV tanpa UPC belum ter-cover. **Fase 2:** Filter by `channel_id` → tampilkan MV tanpa label → staff mapping manual.

Tech Stack & Deployment

Komponen	Pilihan
Framework	Node.js + Fastify

Database	ClickHouse 24.8
Deploy	Satu VM (localhost:3001)
UI	Semua di App A (App B headless)

```
services:
  clickhouse:
    image: clickhouse/clickhouse-server:24.8-alpine
    ports: ["127.0.0.1:8123:8123", "127.0.0.1:9000:9000"]
    volumes: [ch_data:/var/lib/clickhouse]
    deploy: { resources: { limits: { memory: 3G } } }
  analytics-api:
    build: .
    ports: ["127.0.0.1:3001:3001"]
    env_file: .env
```

Resource	Rekomendasi
CPU	8 core
RAM	16 GB
Disk	200 GB NVMe SSD

Peta Jalan

Fase	Fokus
1. Foundation	Setup App B + ClickHouse + DDL + RBAC
2. Ingestion	ETL enrichment + reconciliation + error handling
3. API	7 endpoint analytics + admin CRUD
4. Integrasi	UI upload + dashboard grafik di App A
5. Polish	Logging, job monitor, production config
6. Future	MV anomali report, partner onboarding